PROJECT MILESTONE 1

CLIMATE INTELLIGENT SYSTEM

PROJECT DESCRIPTION

NOVEL NLP DESIGN

In this project we focus on building an end-to-end NLP pipeline that will automatically extract the structured knowledge from unstructured, scientific climate literature. It organizes this into a Knowledge Graph that is queryable.

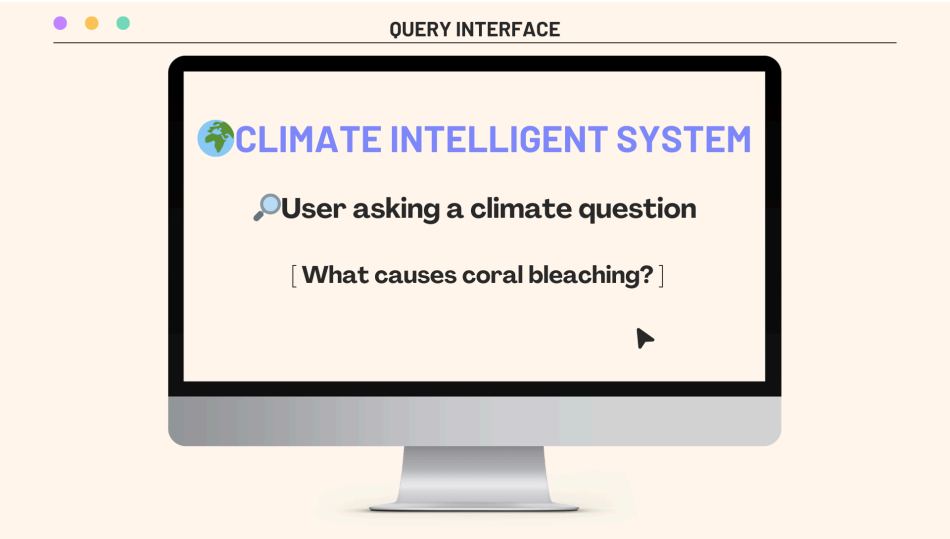
The system basically takes the input which are raw scientific data or abstracts. It then identifies the climate-related entities such as CO2, NO2, glacier melt, deforestation, sea level rise etc. using **Named Entity Recognition**. It then detects relationships between the entities using **Relation Extraction**. The results are extracted as triples which are then stored as edges and nodes in a Climate Knowledge Graph. This knowledge graph will enable graph based analytics like influence ranking, multi-hop impact pathway detection etc.

What the system is meant to do and what use it will have

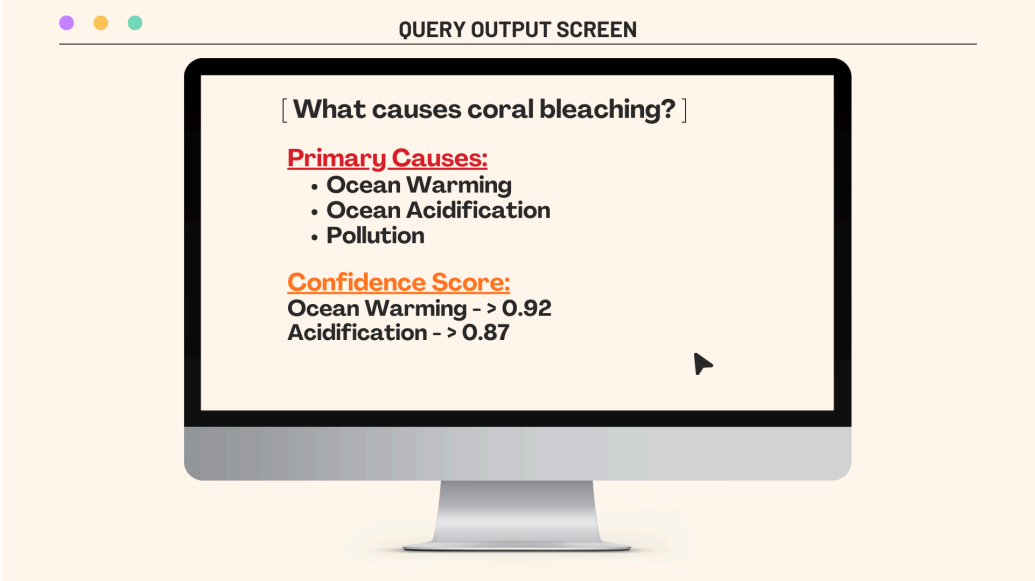
The novel approach of the system is in making climate easily accessible, navigable and computable. This project solves the huge problem of a researcher manually reading hundreds of papers to understand how one climate driver influenced the downstream effects. The Climate Intelligent System allows the researchers to query the system and receive a ranked and evidence chain of consequences. The direct applications of the project could be environmental risk assessment and communication of climate science by transforming thousands of research findings into a single interconnected reasoning engine.

EXAMPLE IMAGES or EXPECTED OUTPUTS

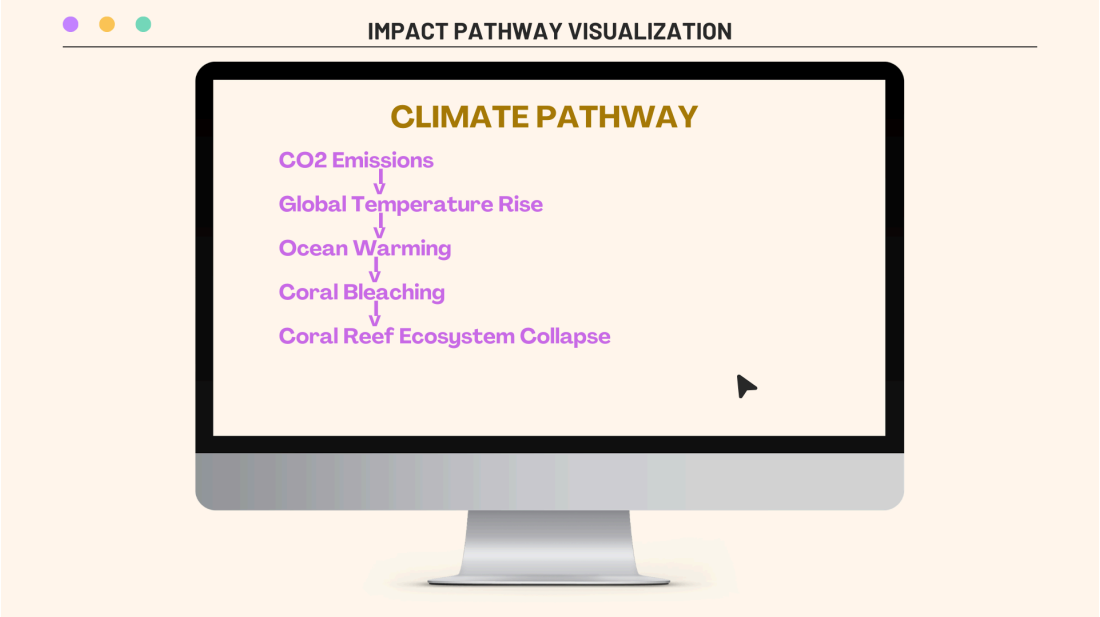
1.



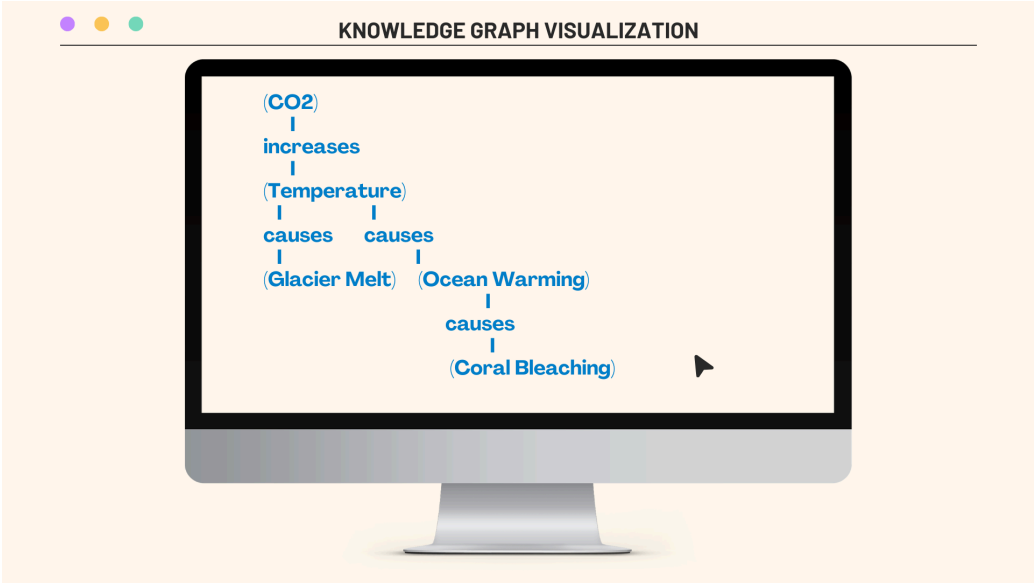
2.


QUERY OUTPUT SCREEN
[What causes coral bleaching?]
Primary Causes:
• Ocean Warming
• Ocean Acidification
• Pollution
Confidence Score:
Ocean Warming - > 0.92
Acidification - > 0.87

3.


IMPACT PATHWAY VISUALIZATION
CLIMATE PATHWAY
CO2 Emissions
Global Temperature Rise
Ocean Warming
Coral Bleaching
Coral Reef Ecosystem Collapse

4.



5.

Data

What type of data is it? What does the data look like (amount, features)? How will you be using the data?

The project would be using scientific climate literature in the form of plain text that consists mainly of primary abstracts and full-text paragraphs. The different data sources that are being considered are:

1. IPCC Reports - These are few in number, authoritative, structured climate science documents.
2. Semantic Scholar API - They are large-scale scientific abstracts accessible via API.

For our initial plan we want to move ahead with the Semantic Scholar API with approximately 500 - 1000 climate related abstracts. The API would be queried using the climate-related keywords like *CO2 emissions, global warming, climate change, coral bleaching* etc. Each retrieved paper provides the following structured fields.

1. Abstract
2. Title
3. Year
4. fieldsofStudy

After annotation each sample will be having the following:

- **Entity spans** labeled across 8 types: Climate Driver, Climate Variable, Environmental Event, Ecosystem, Species, Human Activity, Geographic Location, and Policy.
- **Relation triples** linking entity pairs, e.g., (CO_2, increases, Temperature).

The data will be used for two purposes.

1. **Training and Evaluation:** A labeled subset of approximately **200–300 annotated sentences** will be used to fine-tune both the NER model and the Relation Extraction model, and to evaluate their performance using precision, recall, and F1 score.
2. **Graph Population** : The remaining unannotated abstracts will be processed by the trained pipeline to extract triples at scale, which are then used to construct the Climate Knowledge Graph.

Prompting Strategy

This project does not use few-shot or zero-shot prompting. Both the NER and Relation Extraction components are built on fine-tuned transformer models (SciBERT/BERT), trained on labeled climate text. The Query Engine answers user questions purely through graph traversal and ranking over the constructed Knowledge Graph, no LLM prompting is involved at any stage.

Proposed Method

How will you be using algorithms from class to create your system? Will you need to train a model?

This project is built using a multi-staged NLP pipeline. Where the first stage is **Named Entity Recognition.** The SciBERT model will be fine-tuned to identify and classify climate-related entities from scientific abstracts. SciBERT is chosen over standard BERT because it was pre-trained on a large corpus of scientific text, making it well-suited for domain-specific vocabulary like climate and environmental terminology. The model will be fine-tuned on a labeled dataset of 200–300 annotated sentences using a token classification objective, where

each token is assigned one of 8 entity type labels and a non-entity label. Performance will be evaluated using precision, recall, and F1 score.

The second stage performs **Relation Extraction (RE)**, where another fine-tuned SciBERT model will be trained as a pairwise classifier. Given a sentence and two identified entities, the model predicts the relation between them from a fixed set of 7 relation types: *causes, increases, decreases, affects, contributes_to, occurs_in,* and *mitigates*. The entity pair and surrounding sentence context are fed as input, and the model outputs a relation label, producing structured triples of the form (Entity1, relation, Entity2). These triples form the raw material for the Knowledge Graph.

The third stage constructs and analyzes a **Climate Knowledge Graph** using the extracted triples. Entities become nodes and relations become directed edges, stored and queried using NetworkX. Graph analytics algorithms, specifically Degree Centrality, Betweenness Centrality, and PageRank, are applied to rank the most influential climate drivers and vulnerable ecosystems. At last, a Query Engine is built on top of the graph using graph traversal algorithms to answer user queries such as *"What causes coral bleaching?"* by finding and ranking connected nodes through their incoming and outgoing edges, enabling multi-hop impact pathway discovery without any LLM prompting.

There will be no few/zero shot prompting the model.

RELATED WORK

What previous research are you basing this idea on?

As this project is a combination of NLP, transformer based model fine tuning, the following are the related papers that we found are more relevant towards our work.

1. https://aclanthology.org/D19-1371.pdf
   This one is most relevant to our project. SciBERT is pre-trained on 1.14M scientific papers and 3.1B tokens from Semantic Scholar. We aim to adopt SciBERT as the base model for both our NER and Relation Extraction components, as its scientific pre-training makes it significantly better suited for climate literature than general-purpose BERT.
2. https://ceur-ws.org/Vol-3747/text2kg_paper9.pdf
   This seems to be the closest existing work for our project. It has steps that construct dataset for automatic extraction and structuring climate-change data using relation extraction and NER model. It also has the pipeline design that we thought of implementing in our project.
3. https://arxiv.org/html/2401.06817v1
   This paper shows us the possibility of applying BERT- based NLP to climate literature. This work uses BERT-based NER on over 600,000 climate-focused scientific documents to extract location-specific climate entities and conduct region-specific climate trend

analyses. We can refer to this paper for our approach of using transformer-based NER on climate abstracts.

4. https://arxiv.org/html/2306.02051v3
   This paper provides the theoretical grounding for our Relation Extraction module. It covers pipeline-based RE approaches that perform entity recognition and relation classification sequentially, as well as deep learning techniques for relation extraction.
5. https://ar5iv.labs.arxiv.org/html/2203.14710
   This paper has the architecture that we have proposed for our project. It proposes a NER architecture using SciBERT as the base encoder combined with a word-level interaction layer and a CRF classification layer, treating NER as BIO tag classification and demonstrating competitive results on scientific datasets.
6. https://link.springer.com/article/10.1007/s40747-022-00806-6
   This provides the conceptual basis for our Knowledge Graph construction module. It reviews multiple pipelines for constructing knowledge graphs from scientific literature, including works that fine-tune SciBERT for entity and relation extraction to build structured knowledge graphs from research publications.

Team Members

1. Harshini Domala -
   Data. Responsible for data collection, preprocessing, and Knowledge Graph construction.
2. Mayukha Sista - FE67016
   Models. NER Model, Relation Extraction model, Graph Analytics and Query Engine.